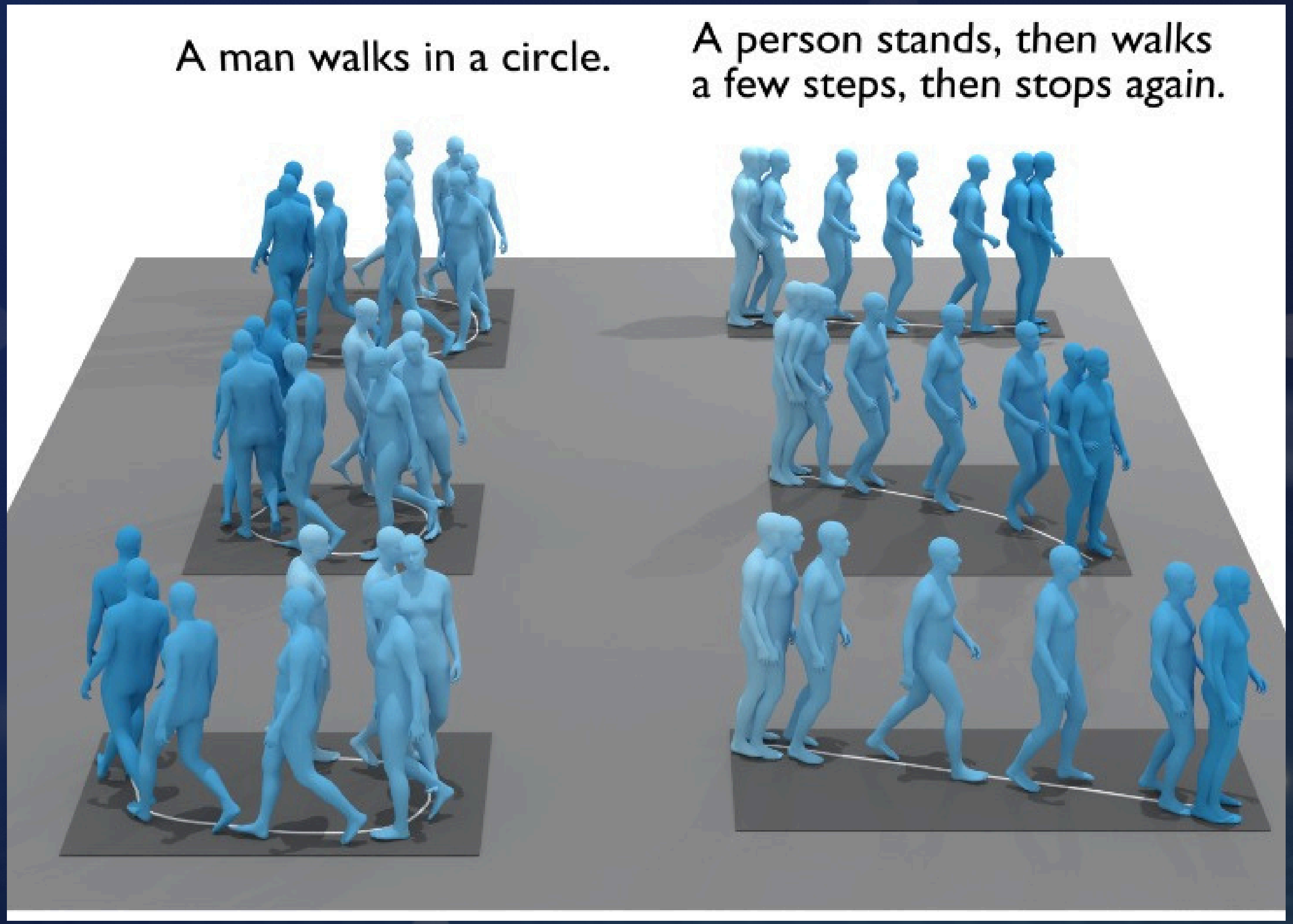
Evaluating Text-to-Motion Models Across Language Backgrounds and Instruction Granularity



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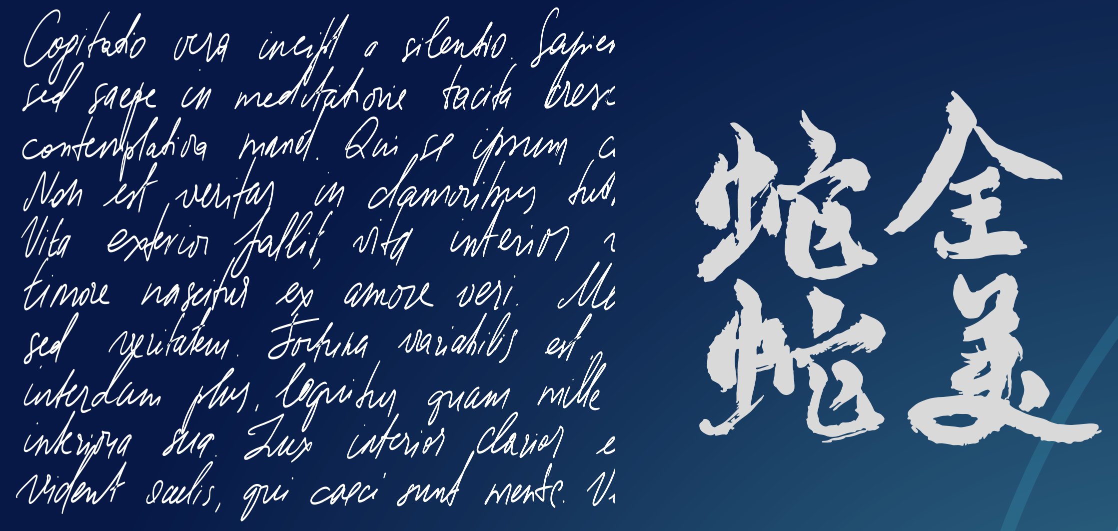
Introduction

- Artificial Intelligence is currently learning to bridge the gap between abstract thought and physical reality

- **Broad Impact:** Advances in text-to-motion have significant implications for several key fields, including robotics, accessibility, and entertainment.

- **Research Gap:** To address gaps in current research, it is necessary to examine how models handle the diversity of human speech.



Problem Statement

The Core Problem:

While natural language is used to control human motion generation, descriptive styles vary drastically between professional and non-professional users.

Linguistic conventions differ significantly across languages (e.g., Chinese vs. English).

Current text-to-motion models are rarely evaluated for their robustness to these linguistic variations.

Project Objectives:

Study how specific linguistic contexts affect motion generation quality.

Analyze the impact of varying instruction details and semantic combinations on model output.

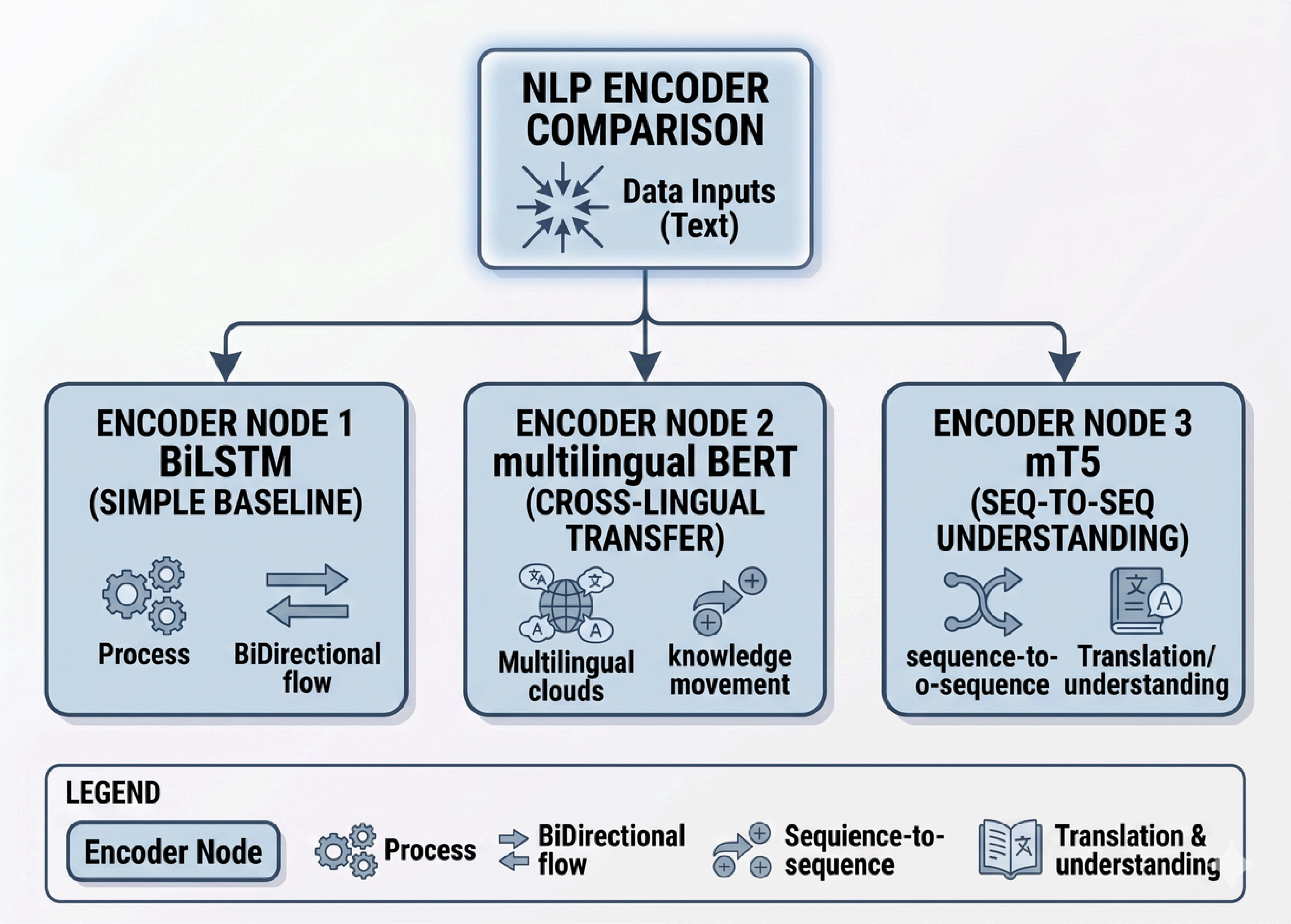
Objective

Goal:
Study how linguistic context affects text-to-motion generation.

Analyze the impact of instruction detail and semantic combinations.

Approach:
Compare several text representation models, including BILSTM, mBERT, and mT5-based text encoders.

Evaluation:
Quantitative retrieval experiments, motion executability assessment, and cross-lingual human evaluation.

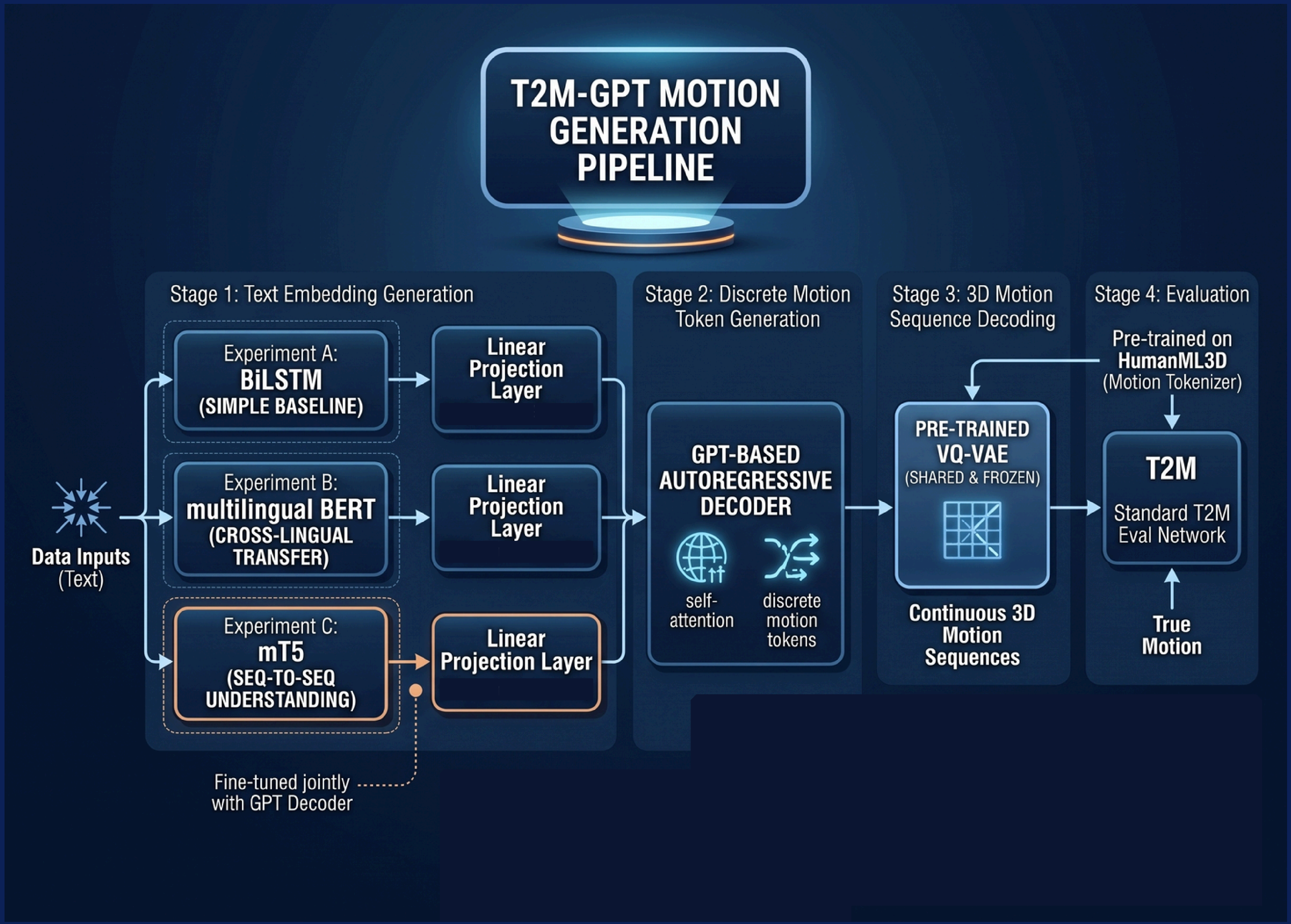


Approach:

We are comparing multiple text representation models, specifically focusing on BILSTM, mBERT, and mT5-based text encoders.

The system takes English text as an input, processes it through the encoder, decodes it using T2M-GPT, and produces a motion sequence.

Dataset: HumanML3D



Method

Experiments

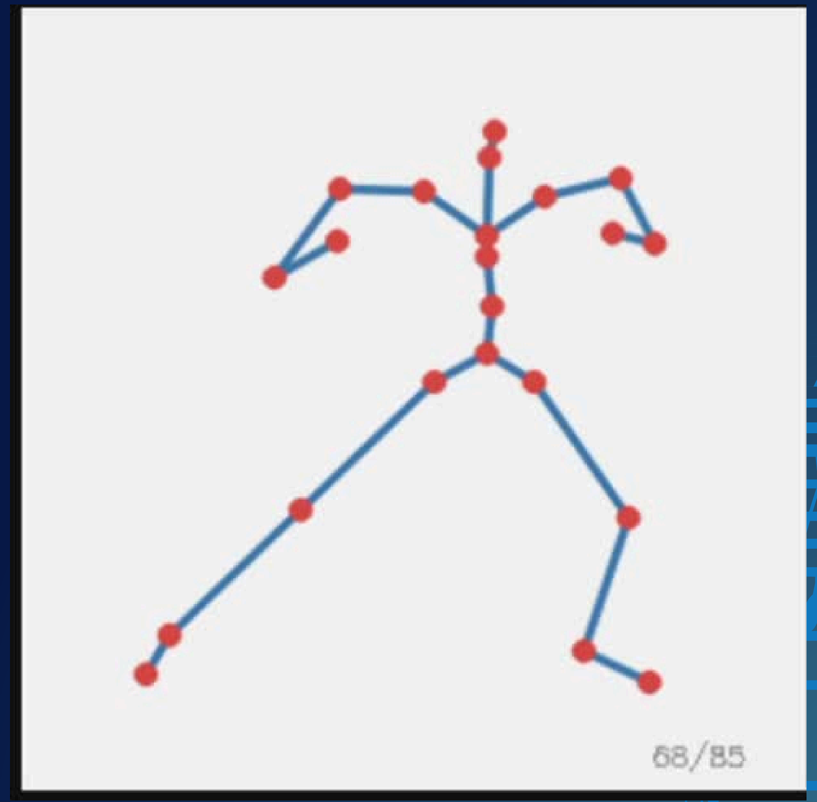
Quantitative Evaluation: R@1, R@5, FID, and MultiModal Distance on existing text-to-motion datasets (e.g., official test split on HumanML3D).

Human Evaluation:

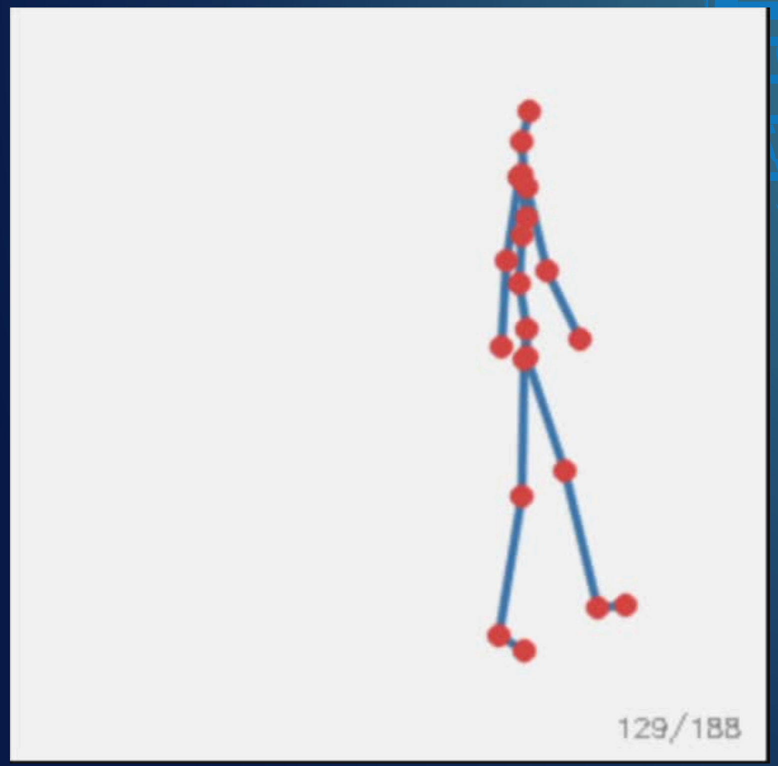
Cross-language: Generate 100 Chinese (zero-shot) and 100 English movement description sentences.

Multi-Granularity: A mixture of simple and complex sentences.

Testing which of three movements best describes the input sentence.



a person slowly punches with their right hand and then slowly punches with their left hand.

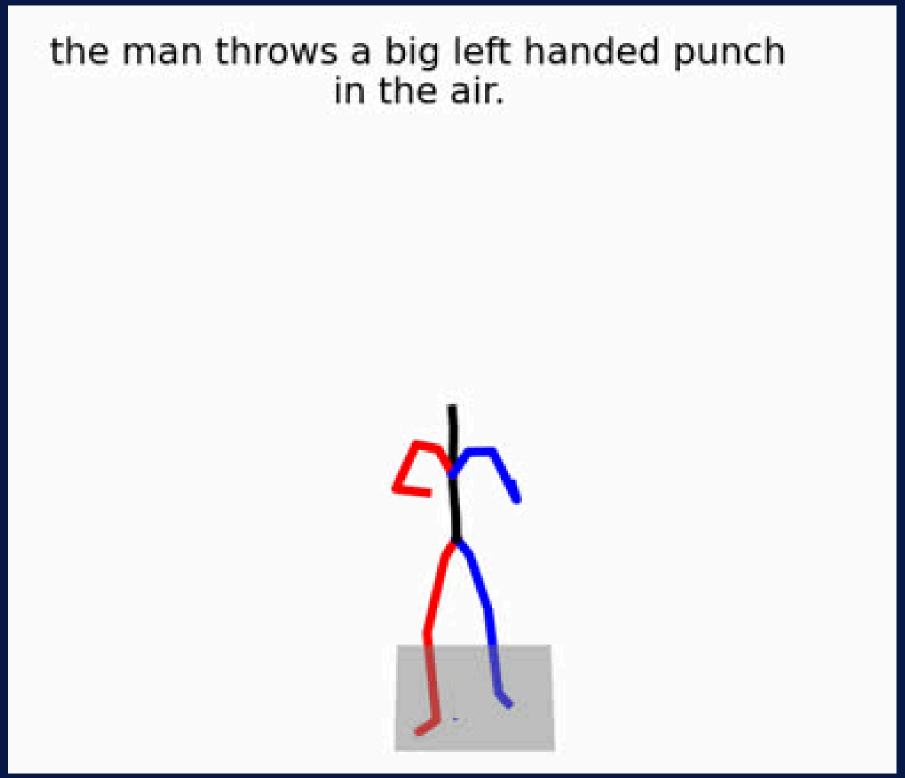


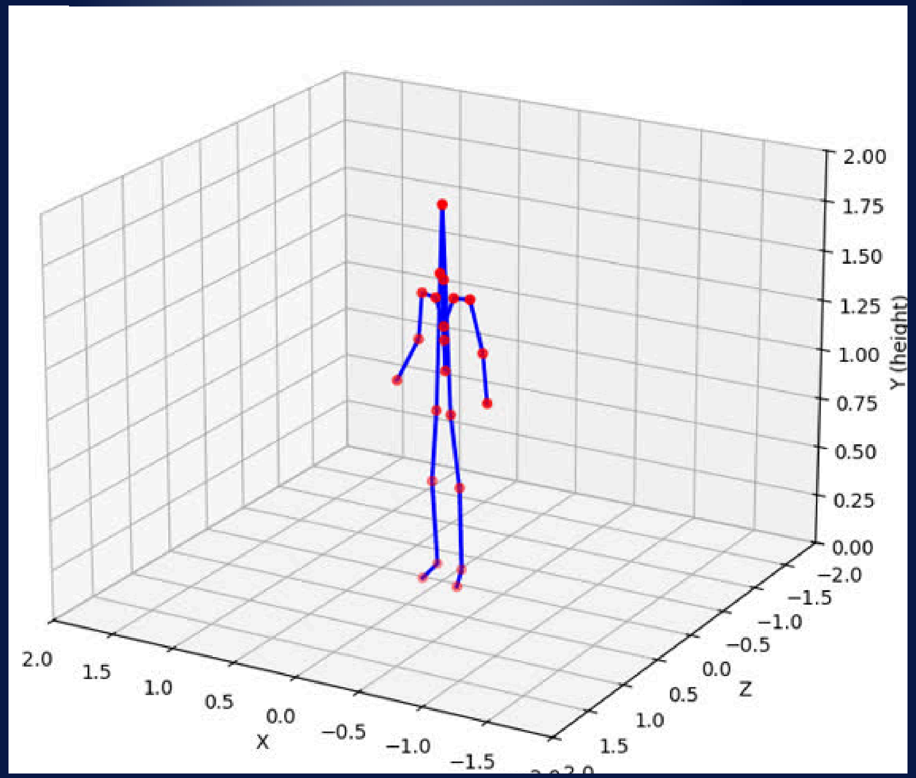
the person paces to his right, then to his left, and finally to his right again.

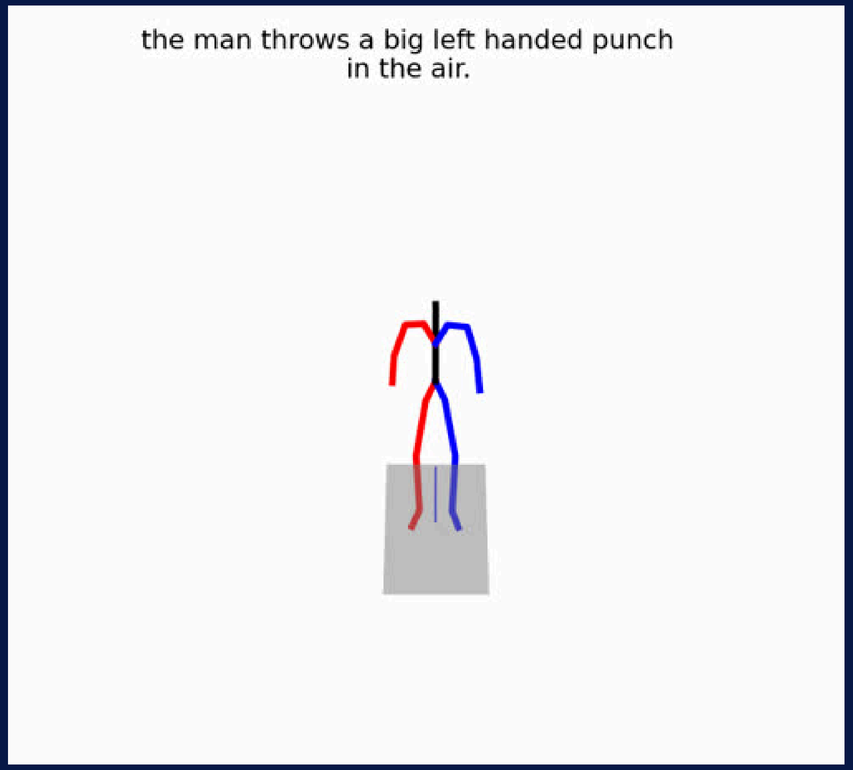
Visualization

	mBert	mT5	BiLSTM

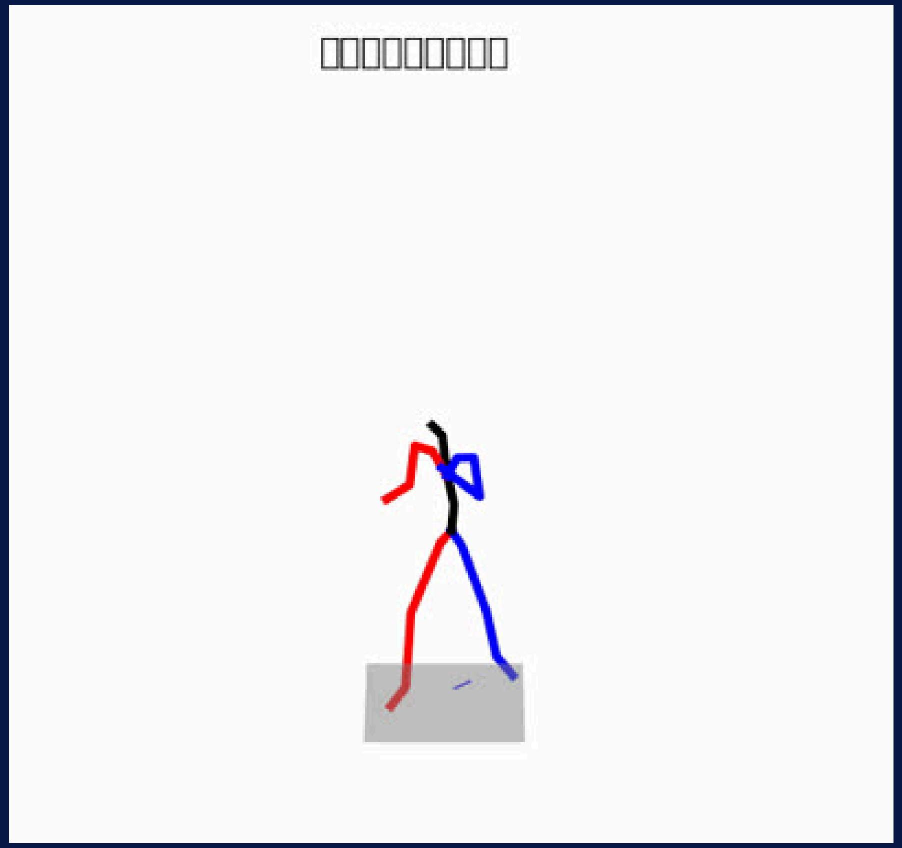
English

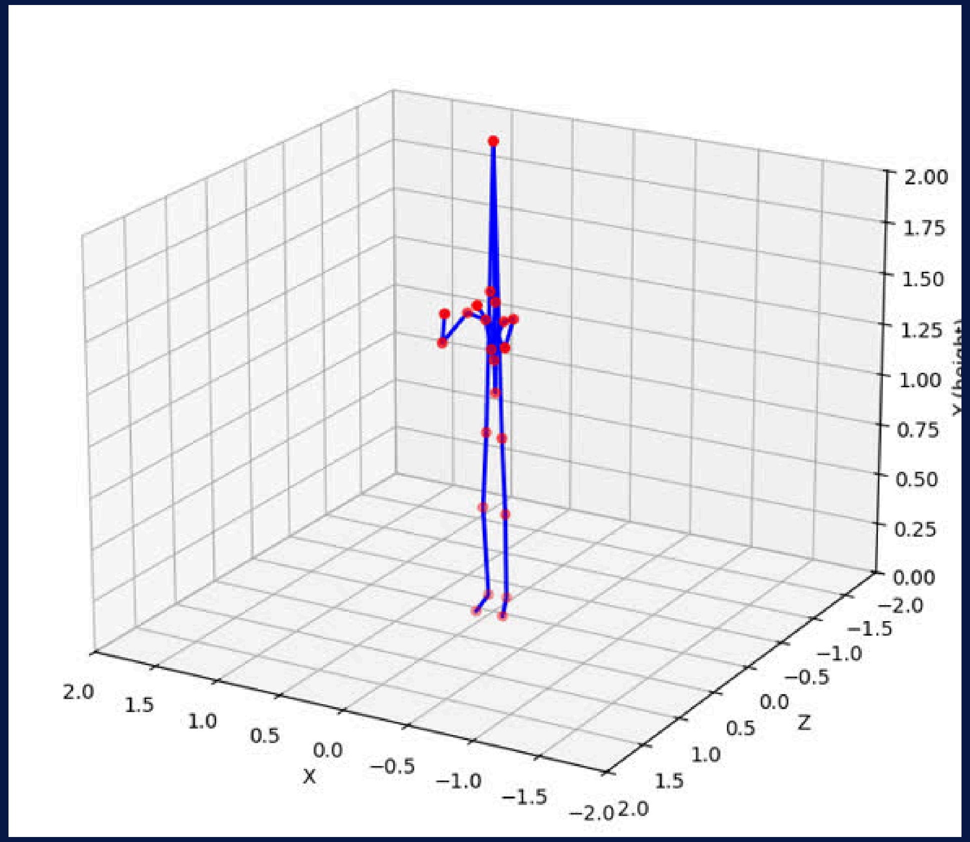


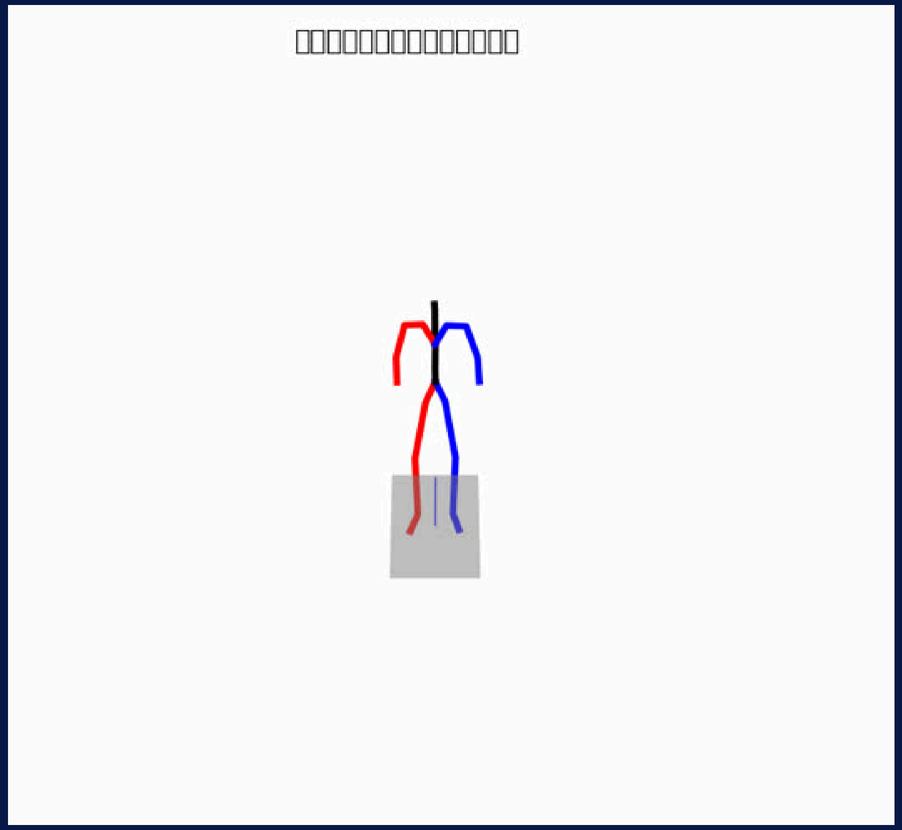




Chinese







Quantitative Metrics

Model	FID	R@1	R@2	R@3	Diversity	Matching Score
BiLSTM	1.724	0.256	0.41	0.528	7.1766	4.21
Multilingual BERT	0.07336	0.5074	0.7176	0.8054	10.2206	2.8552
mT5	0.1857	0.4704	0.6718	0.7754	10.2409	2.9582

Conclusion and Team Contributions

To achieve true cross-lingual motion generation, we must shift from English-centric architectures to multilingual models like mT5 and mBERT. Ultimately, the field must evolve its evaluation standards to prioritize linguistic diversity and real-world robustness over standard English benchmarks.

Team Contributions:

Angie Song (as620): BILSTM fine-tuning and quantitative evaluation.

Serena (Xin) You (xy68): mBERT fine-tuning and quantitative evaluation.

Shizuka Dara (sd205): mT5 fine-tuning and quantitative evaluation.

Tanishka Sonar (ts158): Visualization and human evaluation.

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